

Musk: Third Person Receives Computer-Brain Connection from Neuralink

Businessman Elon Musk said a third person has had a special electronic device directly connected to their brain.

The unidentified person received an implant from Musk's company Neuralink, which makes devices that link the human brain to computers. Neuralink is one of many groups working to connect the human nervous system to machines.

On the social media service X, which he owns, Musk said: "We've got...three humans with Neuralinks and all are working well."

Neuralink announced its first brain implant one year ago. Recently, Musk said the company has improved its devices. Musk also said Neuralink hopes to implant the experimental devices in 20 to 30 more people this year.

FILE - Elon Musk speaks as part of a campaign town hall in support of Republican presidential nominee former President Donald Trump in Folsom, Pa., Thursday, Oct. 17, 2024.

The second patient reportedly has a spinal cord injury and got the implant last summer. The person was playing video games with the help of the device and learning how to use design software. The first patient, also paralyzed after a spinal cord injury, said it helped him play video games and chess.

Many other companies and research groups are working on similar projects. Two studies published last year in the New England Journal of Medicine describe how brain-computer interfaces, known as BCIs, helped people with Amyotrophic lateral sclerosis (ALS) communicate better. ALS is a nervous system disease that affects cells in the brain and spinal cord.

Others working on brain-computer interface technology

Nature magazine reported last September that there were published results from 28 trials worldwide of BCIs over the past 25 years.

Rajesh Rao is co-director of the Center for Neurotechnology at the University of Washington. Rao said that many research laboratories have already shown that humans can control computer cursors using BCIs.

Rao said Neuralink might be different from the others in two ways. First, the surgery to implant the device is the first to use a robot to implant flexible electrical connectors into a human brain

to record brain signals and to control devices. Second, those connectors might record from more parts of the brain than interfaces from other research efforts.

Rao said other companies such as Synchron, Blackrock Neurotech and Onward Medical are carrying out BCI tests on people using methods that are different from Neuralink.

Benefits of BCIs

Marco Baptista is chief scientific officer of the Christopher & Dana Reeve Foundation. He called BCI technology "very exciting" with possible benefits to people with paralysis.

Through clinical tests, "we'll be able to see what's going to be the winning approach," he said. "It's a little early to know." A clinical trial follows established rules on how to design an experiment involving human subjects.

How are BCIs tested and supervised?

Neuralink announced in 2023 that it had received permission from U.S. officials to begin testing its device in people. The Food and Drug Administration (FDA) is responsible for approving the sale of medical devices.

Dr. Rita Redberg studies high-risk devices at the University of California, San Francisco. Redberg said that most medical devices go on the market without clinical trials. However, high-risk devices that are seeking approval from the FDA before they go on the market need what is called an "investigational device exemption."

Neuralink says it has this exemption.

I'm Jill Robbins.

Laura Ungar reported this story for the Associated Press. Jill Robbins adapted it for Learning English.

Vocabulary

implant - n. สิ่งที่ยึดเข้าไปในร่างกายของบุคคลโดยการผ่าตัด

spinal cord - n. กลุ่มเส้นประสาทขนาดใหญ่ที่ทอดผ่านศูนย์กลางของกระดูกสันหลังและนำข้อความระหว่างสมองกับส่วนที่เหลือของร่างกาย

paralyzed - adj. ไม่สามารถเคลื่อนไหวหรือรู้สึกถึงร่างกายได้ทั้งหมดหรือบางส่วน

cursor - n. เครื่องหมายบนหน้าจอคอมพิวเตอร์ที่แสดงตำแหน่งที่กำลังป้อนหรืออ่านข้อมูล

benefit - n. ผลลัพธ์ที่ดีจากการกระทำหรือการรักษา

exemption - n. การได้รับการยกเว้นจากการถูกกำหนดให้ทำในสิ่งที่ผู้อื่นต้องทำ

Reading Comprehension Quiz

1. What significant announcement did Elon Musk make regarding the Neuralink device in the article?

- A. Neuralink is planning to halt all human trials.
- B. A second person has successfully received a brain implant.
- C. A third person has had a special electronic device connected to their brain.
- D. The first two Neuralink implants have failed to function.

2. What is the primary purpose of Neuralink's devices, as stated in the text?

- A. To diagnose nervous system diseases.
- B. To link the human brain to computers.
- C. To develop new communication methods for social media.
- D. To assist in traditional surgical procedures.

3. Based on Elon Musk's statement on the social media service X, what is the current status of the Neuralink implants in the three human subjects?

- A. They are still in the initial testing phase.
- B. Some devices are experiencing technical difficulties.
- C. All three implants are functioning effectively.
- D. The results are inconclusive so far.

4. When did Neuralink first announce its brain implant?

- A. Last summer.
- B. This year.
- C. One year ago.
- D. 25 years ago.

5. How many additional people does Neuralink hope to implant experimental devices in this year?

- A. Less than 10 people.
- B. Between 20 to 30 people.
- C. More than 50 people.
- D. An unspecified number, depending on trial success.

6. What common injury did both the first and second Neuralink patients experience, leading to their paralysis?

- A. A severe head injury.
- B. Amyotrophic lateral sclerosis (ALS).
- C. A spinal cord injury.
- D. A stroke affecting brain function.

7. What activities did the first and second Neuralink patients achieve with the help of the device?

- A. They were able to walk again and regain speech.
- B. They played video games, chess, and used design software.
- C. They could operate heavy machinery and drive vehicles.
- D. They learned new languages and composed music.

8. According to two studies published last year, how did Brain-Computer Interfaces (BCIs) help people with Amyotrophic lateral sclerosis (ALS)?

- A. By restoring their motor functions.
- B. By improving their communication abilities.
- C. By curing their disease entirely.
- D. By preventing the progression of ALS.

9. How many BCI trials had published results worldwide over the past 25 years, as reported by Nature magazine last September?

- A. 2 trials.
- B. 15 trials.
- C. 28 trials.
- D. Over 50 trials.

10. What capability has Rajesh Rao stated that many research laboratories have already demonstrated using BCIs?

- A. Humans can control robotic limbs with precision.
- B. Humans can control computer cursors.
- C. Humans can directly access internet data.
- D. Humans can generate complex thought patterns.

11. According to Rajesh Rao, what is one key difference in Neuralink's surgical procedure compared to others?

- A. It uses conventional surgical instruments.
- B. It is the first to use a robot to implant flexible electrical connectors.
- C. It involves a non-invasive brain scanning technique.

D. It only requires local anesthesia for the patient.

12. What is the second potential difference of Neuralink's connectors identified by Rajesh Rao?

- A. They are designed to be entirely invisible once implanted.
- B. They might record from more parts of the brain than other interfaces.
- C. They are made from organic, biodegradable materials.
- D. They do not require any external power source.

13. Which companies are mentioned by Rajesh Rao as also conducting BCI tests on people with different methods than Neuralink?

- A. Microsoft, Amazon, and IBM.
- B. Synchron, Blackrock Neurotech, and Onward Medical.
- C. Google, Apple, and Samsung.
- D. Tesla, SpaceX, and X.

14. What is Marco Baptista's role and his general sentiment towards BCI technology?

- A. He is a journalist who is wary of its ethical implications.
- B. He is a regulatory official who views it with caution.
- C. He is a chief scientific officer who finds it 'very exciting' with benefits for paralysis.
- D. He is a patient advocate who believes it is progressing too slowly.

15. Which U.S. government agency is responsible for approving the sale of medical devices, as mentioned in the article?

- A. The Department of Health and Human Services (HHS).
- B. The National Institutes of Health (NIH).
- C. The Food and Drug Administration (FDA).
- D. The Centers for Disease Control and Prevention (CDC).